

	Case Name: Sanitary Sewer Network Infrastructure Project Owner: Eder Omar Vilca Carrillo	Sector	Construction	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Processed by	Zahra Hussaini		One of nine std. scenarios	
Date	2025		User defined distributions	
File Name	C2025-22	Project Control	Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project describes the different activities and their interrelations in the construction and improvement of a sanitary sewer system, including excavation, pipe installation, manhole construction, backfilling, safety measures, and final restoration works, aimed at enhancing public sanitation services and ensuring safe wastewater management in the area. The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	45	Serial/Parallel (SP)	38%
Planned Duration (PD)	27 days	Activity Distribution (AD)	29%
Budget At Completion (BAC)	73441.85 €	Length of Arcs (LA)	0%
Renewable Resources	7 Types	Topological Float (TF)	31%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0.0	0.1	6.7
CRI-rho	0.7	0.0	6.7
CRI-tau	1.0	0.1	-6.7

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	0.4	0.5	0.4
SI	0.5	0.4	0.2
SSI	0.1	0.1	0.9
CRI-r	0.1	0.1	1.1
CRI-rho	0.1	0.1	1.2
CRI-tau	0.1	0.1	0.4

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	20.0	-12.2
PV - SPI	31.7	17.4
PV - SCI	31.7	17.4
ED - 1	25.4	-0.7
ED - SPI	31.7	17.4
ED - SCI	31.7	17.4
ES - 1	15.9	-14.9
ES - SPI(t)	12.0	7.7
ES - SCI(t)	12.0	7.7

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0	0
CPI	0	0
SPI	20.7	20.7
SPI(t)	14.3	14.3
SCI	20.7	20.7
SCI(t)	14.3	14.3
0.8 CPI + 0.2 SPI	10.1	10.1
0.8 CPI + 0.2 SPI(t)	4.7	4.7

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 2 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	1.0	-3466.1	-148	1.0	1.0	0.5	1.0
std dev	0.0	4901.8	5.7	0.0	0.1	0.1	0.0
final	1.0	0.0	-152	1.0	1.0	0.6	1.0

2.3.3.2. Time forecasting

PD	27 days	Real Duration	46 days	Late	41%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	27.5	2.1	20.1	20.1
PV - SPI	27.5	2.1	20.1	20.1
PV - SCI	27.5	2.1	20.1	20.1
ED - 1	42	4.2	4.3	4.3
ED - SPI	42	4.2	4.3	4.3
ED - SCI	42	4.2	4.3	4.3
ES - 1	44.5	0.7	1.6	1.6
ES - SPI(t)	49	5.7	3.3	-3.3
ES - SCI(t)	49	5.7	3.3	-3.3

2.3.3.3. Cost forecasting

BAC	73441.85 €	Real Cost	73441.85 €	On budget	0%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	73441.9	0.0	1E-05	-1E-05
CPI	73441.8	0.1	7E-05	6E-05
SPI	73803.1	510.9	0.5	-0.5
SPI(t)	76908.0	4901.8	4.7	-4.7
SCI	73803.1	510.8	0.5	-0.5
SCI(t)	76907.9	4901.7	4.7	-4.7
0.8 CPI + 0.2 SPI	73508.5	94.3	0.1	-0.1
0.8 CPI + 0.2 SPI(t)	73826.9	544.6	0.5	-0.5